

FOOD & HEALTH NUTRITION ANALYTICS

Phase I — Beginner SQL Exercises (1–20)

Questions Only — Student Worksheet

Part A — Basic SELECT and Filtering

1. Display all columns and all records from the Foods table.
 2. Display only the FoodID, FoodName, CaloriesPer100g, ProteinPer100g, and FiberPer100g columns from the Foods table.
 3. Display all food items that contain more than 10 grams of protein per 100 grams.
 4. Display all food items that contain at least 5 grams of fiber per 100 grams.
 5. Find all food items that contain less than 100 calories per 100 grams.
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Part B — Sorting and Limiting Results

6. Display all food items in descending order of protein content, with the highest-protein food appearing first.
 7. Display all food items in ascending order of calories per 100 grams, with the lowest-calorie food appearing first.
 8. Display the 5 foods with the highest protein content.
 9. Display the 10 foods with the highest fiber content.
 10. Display the 5 foods with the lowest calorie content.
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Part C — Searching and Filtering

11. Find all food items whose FoodName contains the word "Rice".
 12. Find all food items whose FoodName starts with the letter "C".
 13. Find all food items belonging to a particular CategoryID. Use CategoryID = 3 for this exercise.
 14. Find all food items whose protein content is between 5 and 15 grams per 100 grams.
 15. Find all food items that have both protein of at least 10 grams and fiber of at least 5 grams per 100 grams.
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Part D — Aggregate Functions

16. Find the total number of food items available in the Foods table.
17. Find the average calorie content of all food items.
18. Find the highest protein content among all food items.
19. Find the lowest fiber content among all food items.
20. — Final Phase I Challenge:
Find the average calories, average protein, and average fiber of all food items. Display the results with meaningful column names such as `AverageCalories`, `AverageProtein`, and `AverageFiber`.

Phase II — Intermediate SQL Exercises (21–45)

Part A — JOIN Operations

21. Display each food item together with its food category name. Display the `FoodName` and `CategoryName`.
 22. Display the `FoodName`, `CategoryName`, `CaloriesPer100g`, `ProteinPer100g`, and `FiberPer100g` for every food item.
 23. Find all foods containing more than 10 grams of protein per 100 grams. Display the `FoodName`, `CategoryName`, and `ProteinPer100g`.
 24. Display every food together with its nutrient name and nutrient amount per 100 grams. Display the `FoodName`, `NutrientName`, and `AmountPer100g`.
 25. Find all foods that contain Iron. Display the `FoodName`, `NutrientName`, and amount of Iron per 100 grams.
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Part B — Foods and Health Conditions

26. Display all food items associated with health conditions. Display the `FoodName`, `ConditionName`, `SuitabilityScore`, and `EvidenceLevel`.
27. Find all foods associated with Type 2 Diabetes. Display the `FoodName`, `Dietary Goal`, `SuitabilityScore`, and `EvidenceLevel`.
28. Find all foods associated with Bone Health. Display the `FoodName`, `Dietary Goal`, `SuitabilityScore`, and `EvidenceLevel`.
29. Find all food-health-condition combinations having a `SuitabilityScore` of at least 85. Display the `FoodName`, `ConditionName`, `SuitabilityScore`, and `EvidenceLevel`.

30. Find all foods associated with the dietary goal "Increase Dietary Fiber". Display the FoodName, GoalName, and SuitabilityScore.

Part C — GROUP BY and HAVING

31. For each health condition, calculate the number of different food items associated with it. Display the ConditionName and FoodCount.

32. Find health conditions that have at least 5 different food items associated with them. Display the ConditionName and FoodCount.

33. For each health condition, calculate the average SuitabilityScore of its associated foods. Display the ConditionName and AverageSuitabilityScore.

34. Find health conditions whose average SuitabilityScore is greater than 80. Display the ConditionName and AverageSuitabilityScore.

35. For each dietary goal, calculate the number of different food items associated with that goal. Display the GoalName and FoodCount.

Part D — CASE Expressions

36. Classify each food according to its protein content per 100 grams:

- 15 grams or more → Very High Protein
- 10–14.99 grams → High Protein
- 5–9.99 grams → Moderate Protein
- Below 5 grams → Low Protein

Display the FoodName, ProteinPer100g, and ProteinCategory.

37. Classify each food according to its fiber content per 100 grams:

- 8 grams or more → Very High Fiber
- 5–7.99 grams → High Fiber
- 3–4.99 grams → Moderate Fiber
- Below 3 grams → Low Fiber

Display the FoodName, FiberPer100g, and FiberCategory.

38. Create a NutritionProfile for each food using these rules:

- Protein ≥ 10 and Fiber ≥ 5 → High Protein & Fiber

- Protein $\geq 10 \rightarrow$ High Protein
- Fiber $\geq 5 \rightarrow$ High Fiber
- Otherwise $\rightarrow$ Other

Display the FoodName, ProteinPer100g, FiberPer100g, and NutritionProfile.

Part E — Subqueries

39. Find all foods whose protein content is higher than the average protein content of all foods. Display the FoodName and ProteinPer100g.

40. Find all foods whose fiber content is higher than the average fiber content of all foods. Display the FoodName and FiberPer100g.

41. Find all foods whose calorie content is higher than the average calorie content of all foods. Display the FoodName and CaloriesPer100g.

Part F — Ranking and Window Functions

42. Rank all foods according to their protein content, with the food having the highest protein content receiving Rank 1. Display the FoodName, ProteinPer100g, and ProteinRank.

43. Display the RANK(), DENSE_RANK(), and ROW_NUMBER() for every food based on ProteinPer100g. Display the FoodName, ProteinPer100g, ProteinRank, ProteinDenseRank, and ProteinRowNumber.

44. Rank foods separately within each food category according to their protein content. Display the FoodName, CategoryName, ProteinPer100g, and CategoryProteinRank.

45. — Final Phase II Challenge: For every food category, find the top 3 foods with the highest protein content. Display the CategoryName, FoodName, ProteinPer100g, and ProteinRank.

Phase III — Intermediate SQL Exercises (46–75)

Questions Only — Student Worksheet

Part A — Advanced JOIN Analysis

46. Find all food items that are **not currently mapped to any health condition**. Display the FoodID and FoodName.

47. Find all **health conditions that do not have any food items mapped to them**. Display the ConditionID and ConditionName.

48. Display **every food item** and the **number of health conditions** associated with each food. Foods without any health-condition mapping should also be displayed with a count of zero.

49. Find the food items that are associated with **at least 3 different health conditions**. Display the FoodName and number of associated health conditions.

50. Find food items that have a **SuitabilityScore of 80 or higher for at least 2 different health conditions**. Display the FoodName and number of qualifying health conditions.

Part B — Nutrition Analysis

51. Find all foods containing **at least 10 grams of protein per 100 grams**. Display the FoodName, CategoryName, and ProteinPer100g.

52. For each food category, calculate the **average calories, average protein, and average fiber per 100 grams**.

53. Which food category has the **highest average protein content**? Display the CategoryName and average protein.

54. Find all foods that contain **at least 10 grams of protein and at least 5 grams of fiber per 100 grams**. Display the FoodName, CategoryName, ProteinPer100g, and FiberPer100g.

55. Classify every food according to its calorie content as follows:

- Below 100 → Low Calorie
- 100–199 → Moderate Calorie
- 200–299 → High Calorie
- 300 or more → Very High Calorie

Display the FoodName, CaloriesPer100g, and the calculated CalorieCategory.

Part C — Health Condition Analysis

56. Calculate the **average SuitabilityScore for each health condition**. Display the ConditionName and average score, starting with the highest average score.

57. Which health condition has the **highest average SuitabilityScore**? Display the ConditionName and average score.

58. Find health conditions that have **at least 5 different foods with a SuitabilityScore of 80 or higher**. Display the ConditionName and number of qualifying foods.

59. For **each health condition**, find the food having the **highest SuitabilityScore**. Display the ConditionName, FoodName, and SuitabilityScore.

60. For **each health condition**, find the **top 3 foods based on SuitabilityScore**. Display the ConditionName, FoodName, SuitabilityScore, and Rank.

Part D — CTE-Based Analysis

61. Using a CTE, calculate the **average protein content for each food category**. Display the CategoryName and average protein.

62. Find foods whose **protein content is higher than the average protein content of their own food category**. Display the FoodName, CategoryName, ProteinPer100g, and category average protein.

63. Find foods whose **protein content is above the overall average protein** and whose **fiber content is also above the overall average fiber**. Display the FoodName, ProteinPer100g, and FiberPer100g.

64. Find food categories whose **average protein and average fiber are both higher than the overall food averages**. Display the CategoryName, average protein, and average fiber.

65. Calculate an educational **NutritionScore** for every food using:

NutritionScore =

(ProteinPer100g × 2)

+ (FiberPer100g × 3)

- (CaloriesPer100g × 0.10)

Display the FoodName and calculated NutritionScore from highest to lowest.

Part E — Ranking and Window Functions

66. Rank all foods **within each food category** according to their fiber content, with the highest-fiber food receiving Rank 1.

67. Find the **top 2 fiber-rich foods in each food category**. Display the CategoryName, FoodName, FiberPer100g, and Rank.

68. Calculate the average SuitabilityScore for every health condition and **rank the health conditions from highest to lowest average score**.

- 69.** Sort all foods by protein content from highest to lowest. For each food, display its protein content and the **protein content of the previous food** in the ordered list.
- 70.** Sort all foods by protein content from highest to lowest. Display each food's protein content and calculate the **difference between its protein content and the previous food's protein content**.
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Part F — Final Analytical Challenges

- 71.** For every food category, identify the **single food with the highest protein content**. Display the CategoryName, FoodName, and ProteinPer100g.
- 72.** For every dietary goal, identify the **food with the highest SuitabilityScore**. Display the GoalName, FoodName, and SuitabilityScore.
- 73.** Find all foods whose **protein content is higher than the average protein content of their own category**. Display the FoodName, CategoryName, ProteinPer100g, category average protein, and difference from the category average.
- 74.** Identify the **10 food-condition combinations with the highest SuitabilityScore**. Display the FoodName, ConditionName, SuitabilityScore, and EvidenceLevel.
- 75. — Final Challenge:** Create a consolidated **Food Health Analytics Report** containing the following information for every food:
- FoodName
 - CategoryName
 - CaloriesPer100g
 - ProteinPer100g
 - FiberPer100g
 - Number of associated health conditions
 - Average SuitabilityScore
 - Protein rank within its food category

For the MySQL case study with 8–10 normalized tables, we can create an Excel workbook like:

Sheet	Purpose
Foods	100 food items
Food_Categories	Food category master
Nutrients	20 nutrients
Food_Nutrients	Food–nutrient relationships
Health_Conditions	15 health conditions
Dietary_Goals	Dietary goals
Food_Health_Benefits	Food–health-condition relationships
Serving_Sizes	Serving-size information
Evidence_Sources	Nutrition/evidence references
Food_Alternatives	Alternative/substitute foods

This would allow students to see the relationship:

